Name_____ Date_____ Period_____

LESSON 9-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Independent and Dependent Variables

Maryland Standard 6.EE.C.9 · Learning target – I can name the independent and dependent variables in a relationship.

REMEMBER

- The **independent variable** is the input you choose or control.
- The **dependent variable** is the output. It **depends on** the input.
- On a graph: independent on the **x-axis**, dependent on the **y-axis**.

WORKED EXAMPLE

The more cars Andre washes, the more money he earns. Name the variables.

Step 1 Andre controls how many cars he washes.

Step 2 Test: "Money earned depends on cars washed." It makes sense.

Answer **Independent: cars washed · Dependent: money earned**

WORD BANK

independent variable – the input; what you choose or control

dependent variable – the output; it changes because of the input

depends on – changes because of

x-axis – the line across the bottom of a graph

PRACTICE

show your work · label units

1. For each pair of quantities, write **I** under the independent variable and **D** under the dependent variable.

Step 1 Read each pair of quantities.

Step 2 Ask: which one changes because of the other? Write D under it.

Step 3 Write I under the other one.

	Quantity	I or D	Quantity	I or D
a.	minutes biking		miles traveled	
b.	total cost		tickets bought	
c.	guests at a party		cups of juice needed	

2. **Bus Ride.** The longer Imani rides the bus, the farther she gets from home. Write a sentence about these two quantities that uses "depends on."

Frame ___ *depends on* ___.

Sentence_____

3. **Dog Walking.** Rosa earns **$6** for each dog she walks. Name both variables.

   Step 1 Ask: what does Rosa choose?

   Step 2 Ask: what changes because of that choice?

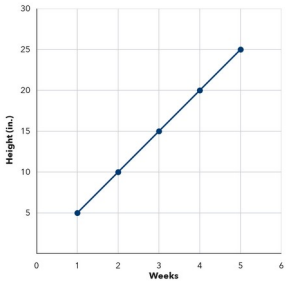
   Independent / Dependent_____

4. **Sunflower.** Kenji graphs the height of his sunflower each week.

   a. Name the independent variable. _____

   b. Name the dependent variable. _____

   c. What does the point (4, 20) mean? _____



5. **Find the Error.** The number of people at a picnic changes how many sandwiches are needed. Leah says the sandwiches are independent because that is the bigger number. Explain her mistake.

   Frame *Leah is wrong because ___. The independent variable is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Name two things at home that change together. Which one depends on the other?

Name_____ Date_____ Period_____

LESSON 9-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Tables of Values

Maryland Standard 6.EE.C.9 · Learning target – I can build a table of values for a relationship between two variables.

REMEMBER

- A **table of values** pairs each input (x) with its output (y).
- Use the **rule** on every input. With a fee, **multiply first**, then add.

WORKED EXAMPLE

A copier prints 20 pages each minute. Make a table for 1, 2, and 4 minutes.

Step 1 Minutes is x. Pages is y. The rule is $y = 20x$.

Step 2 $20 \times 1 = 20$, $20 \times 2 = 40$, $20 \times 4 = 80$

Answer **20, 40, and 80 pages**

WORD BANK

table of values – a list of inputs and their outputs

rule – what you do to each input

fee – a cost you pay one time

input – the number you start with (x)

PRACTICE

show your work · label units

1. Use the rule $y = x + 8$ to complete the table.

Step 1 Put each x into the rule.

Step 2 Add 8 to get y.

x	0	2	4	7
y				

2. **Skate Rental.** A rink charges $5 to get in plus $3 for each hour of skating. Complete the table.

Step 1 Multiply the hours by $3.

Step 2 Then add the $5 to get in.

Hours (*x*)	1	2	3	6
Cost in $ (*y*)				

3. Marker Boxes. The table shows markers in boxes. Describe the rule. Then fill in the missing values.

Boxes (x)	1	2	3	4	8
Markers (y)	8	16	24		

Rule_____

4. Find the Error. Marcus made a table for $y = 2x + 1$. He wrote x: 1, 2, 3, 4 and y: 3, 5, 7, 8. Which y value is wrong? What should it be?

Frame *The wrong value is ___ because 2 × 4 + 1 = ___.*

Answer_____

5. Stretch. A taxi charges $4 plus $2 for each mile. Complete the table. How many miles cost $20?

Step 1 Multiply the miles by $2, then add $4.

Step 2 For $20: take away the $4, then divide by 2.

Miles (x)	1	3	5	
Cost in $ (y)				20

TEST PRACTICE

choose the best answer

6. A rule is "multiply by 4, then add 5." What is y when $x = 6$?

A 24 **B** 44 **C** 29 **D** 15

7. A table has x-values 1, 2, and 3. Which y-values match the rule $y = x + 6$?

A 6, 12, 18 **B** 7, 8, 9 **C** 7, 14, 21 **D** 6, 7, 8

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___